

What are Angular lifecycle hooks?

Lifecycle hooks are special functions in Angular that let you tap into different phases of a component or directive's life:

- `ngOnInit()`: Called after the constructor and the first `ngOnChanges()`.
- `ngOnChanges()`: Called when input-bound properties change.
- `ngOnDestroy()`: Called before the component is destroyed.
- Others: `ngDoCheck()`, `ngAfterViewInit()`, `ngAfterContentInit()`, etc.

Difference between `ngOnInit()` and constructor?

- constructor: Used for dependency injection and basic initialization.
- `ngOnInit()`: Called after inputs are initialized; used for complex setup logic.

What is the difference between `@Input()` and `@Output()`?

- `@Input()`: Allows parent component to send data to child.
- `@Output()`: Allows child to send data/events to parent using `EventEmitter`.

What is the purpose of `ngOnDestroy()`?

Used to clean up resources (subscriptions, timers, DOM listeners) before a component is destroyed.

What are pipes in Angular?

Pipes transform data in templates. Example: `{{ price | currency:'INR' }}`. Custom pipes can be created using `ng generate pipe`.

What is a reactive form vs template-driven form?

Template-driven: HTML-based, uses `ngModel`.

Reactive: Component class-based using `FormControl` and `FormGroup`.

Reactive forms are more scalable and testable.

What is `HttpClient` and how do you use it?

Used to make HTTP requests.

Example:

```
this.http.get('https://api.example.com/data');
```

Import `HttpClientModule` in `AppModule`.

How do you share data between components?

- Parent to child: `@Input()`
- Child to parent: `@Output()`
- Siblings: Shared service with `Subject` or `BehaviorSubject`.

What is lazy loading in Angular?

Feature modules are loaded only when needed to improve performance.

Implemented via route configuration with `loadChildren`.

What are services and how are they injected?

Services share logic/data. Injected using dependency injection via constructors. Use @Injectable({ providedIn: 'root' }).

What is a Subject and BehaviorSubject?

- Subject: No initial value, emits to multiple subscribers.
- BehaviorSubject: Has an initial value, emits latest value to new subscribers.

What is the role of NgModule?

Defines a context for components, services, pipes, and directives. Tells Angular what is available in the application.

What is Change Detection in Angular?

It updates the DOM when data-bound properties change. Optimize using OnPush strategy and trackBy in ngFor.

What is Routing Guard in Angular?

Controls navigation using guards like CanActivate, CanDeactivate, CanLoad.

What is the difference between ngIf and *ngIf?

*ngIf is syntactic sugar for structural directive ngIf. It expands to for DOM manipulation.